电子科技大学 格拉斯哥 学院

UOG-UESTC Joint School of UESTC

标 准 实 验 报 告

Lab Report

（实验）课程名称： 信号与系统

(LAB) Course Name： Signals and Systems

电子科技大学教务处制表

UoG-UESTC Joint School
Lab Report

Student Name： Chen YiChi **Student No.** [redacted]

Instructor： Aynul Islam

Location： No.6 Scientific Research Build **Date： 2024.10.28**

一、 Laboratory name：

Signals and Systems

二、 Project name：

Represent signals using MATLAB

三、 Lab hours： 4

四、 Reference book:

John R.Buck, Michael M. Daniel, Andrew C. Singer. Computer Exploration in Signals and Systems —— Using MATLAB.

五、 Objective:

1. Be further familiar with the properties of linear time-invariant (LTI) systems.
2. Perform convolution using MATLAB.
3. Compute the output of causal LTI system characterized by linear constant-coefficient differential/ difference equations.

六、 Related commands:

Conv, filter, lsim

七、 MATLAB exercises:

1. Perform convolution. 2.1(a)(b)(c), 2.7(a)(b)(c)(d)(e)
2. Compute the the output of causal LTI system characterized by linear constant-coefficient difference equations . 2.2(a)(b)(c)(d)(e)(f)(g)
3. Compute the the output of causal LTI system characterized by linear constant-coefficient difference equations . 2.3(a)(b)
4. Further familiar with the properties of LTI systems. 2.4(a)(d), 2.5(d)(e)(f)(g

八、 Required instruments：

Computer, MATLAB

九、 Procedures, Analysis of Lab data & result and Conclusion：

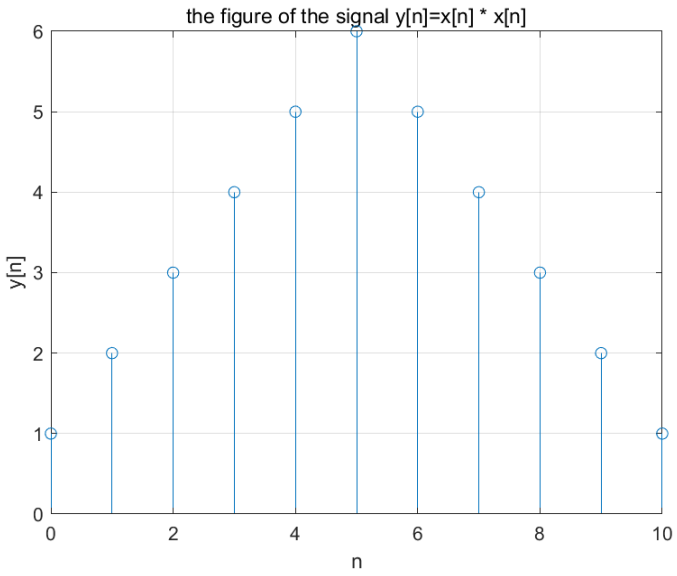
2.1

b

Codes:

```
clc
clear
x=[ones(1,6)];
y=conv(x,x);
ny=[0:length(y)-1];%L=6+6-1-1
stem(ny,y);
xlabel('n');
ylabel('y[n]');
title('the figure of the signal y[n]=x[n] * x[n]');
grid on;
```

Figure:



Explanations:

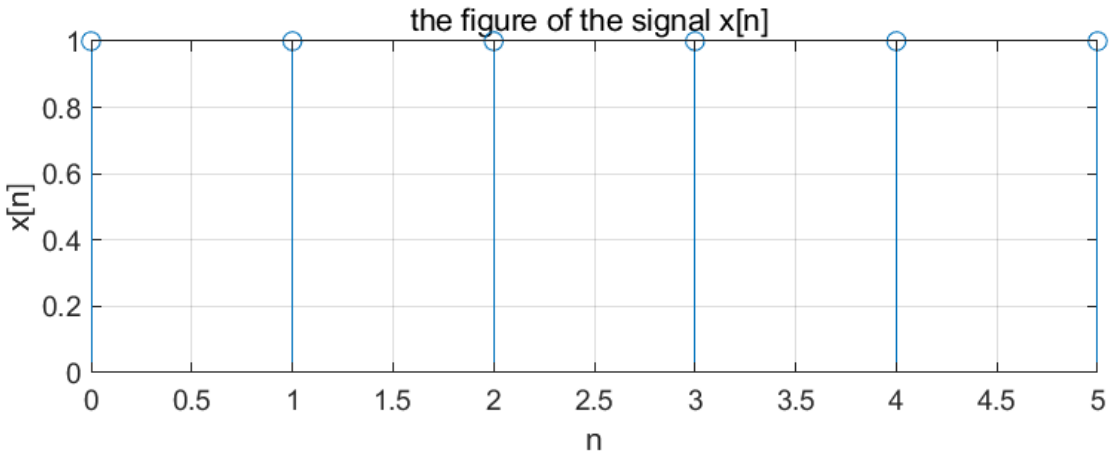
The generated figure was actually the same as the figure in 2.1.a, which prove ny is ranging from nxmin+nxmin to nxmax+nxmax.

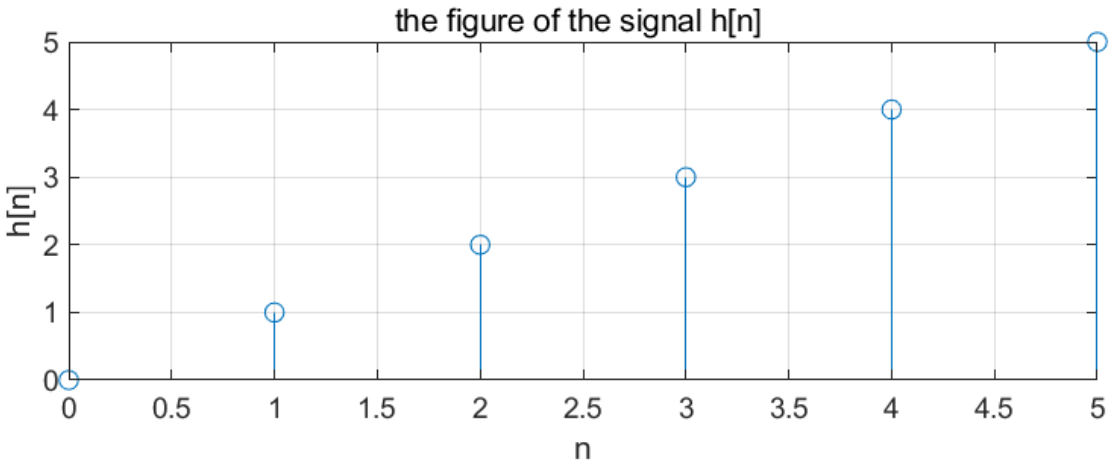
D

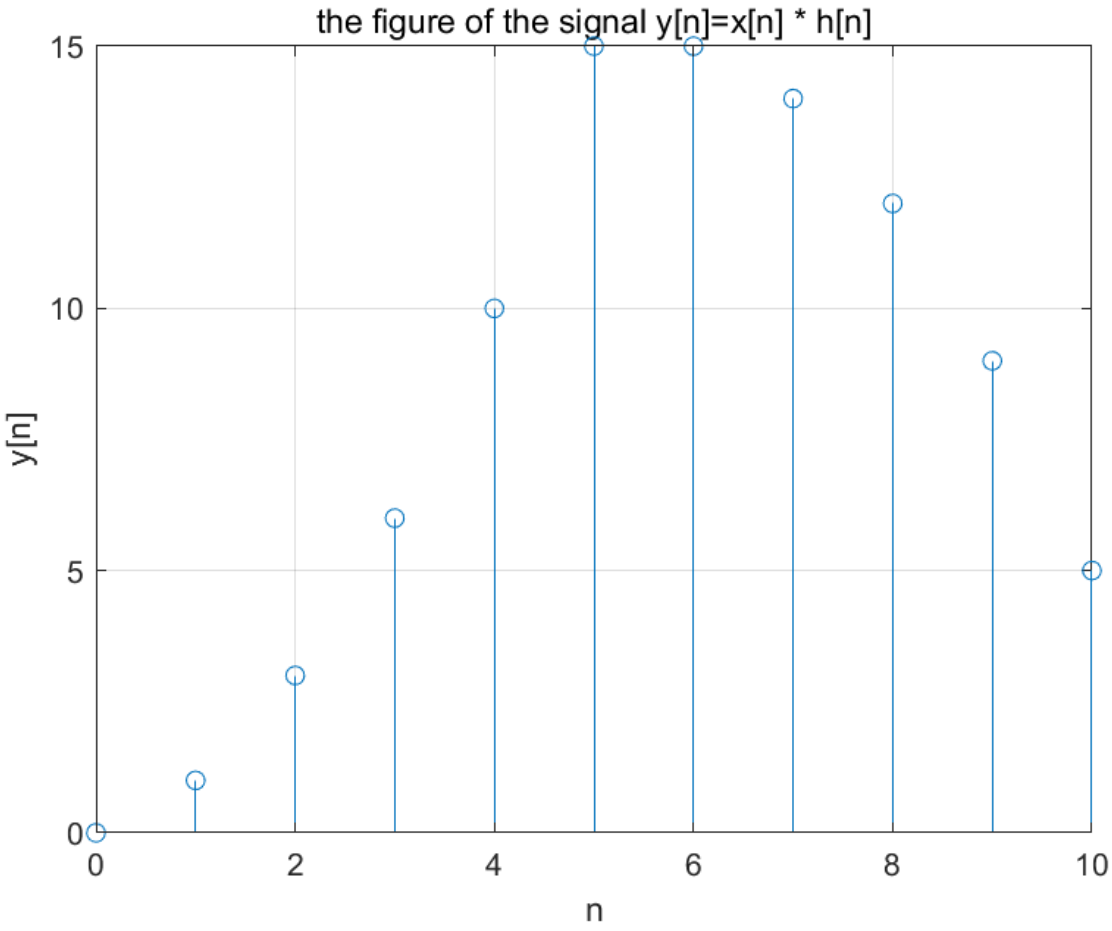
Codes

```
clc
clear
x=[ones(1,6)];
nx=[0:5];
h=[0:5];
nh=[0:5];
y=conv(h,x);
nx=[0:length(x)-1];
nh=[0:length(h)-1];
ny=[0:length(y)-1];
figure(1);
subplot(2,1,1);
stem(nx,x);
xlabel('n');
ylabel('x[n]');
title('the figure of the signal x[n] ');
grid on;
subplot(2,1,2);
stem(nh,h);
xlabel('n');
ylabel('h[n]');
title('the figure of the signal h[n] ');
grid on;
figure(2);
stem(ny,y);
xlabel('n');
ylabel('y[n]');
title('the figure of the signal y[n]=x[n] * h[n]');
grid on;
```

figure







Explanation

From the figure, we can get ny is from 0 to 10,which is equal to nxmin+nhmin and nxmax+nhmax(when X and X do not equal to 0)

2.2.abcd

Codes:

```
clc
clear
```

```
a1=1;
b1=[0.5 1 2];
a2=[1 -0.8];
b2=2;
a3=[1 -0.8];
b3=[0 2];
x=[1 2 3 4];
y1=filter(b1,a1,x)
y2=filter(b2,a2,x)
y3=filter(b3,a3,x)
```

Output:

y1 =

0.5000 2.0000 5.5000 9.0000

y2 =

2.0000 5.6000 10.4800 16.3840

y3 =

0 2.0000 5.6000 10.4800

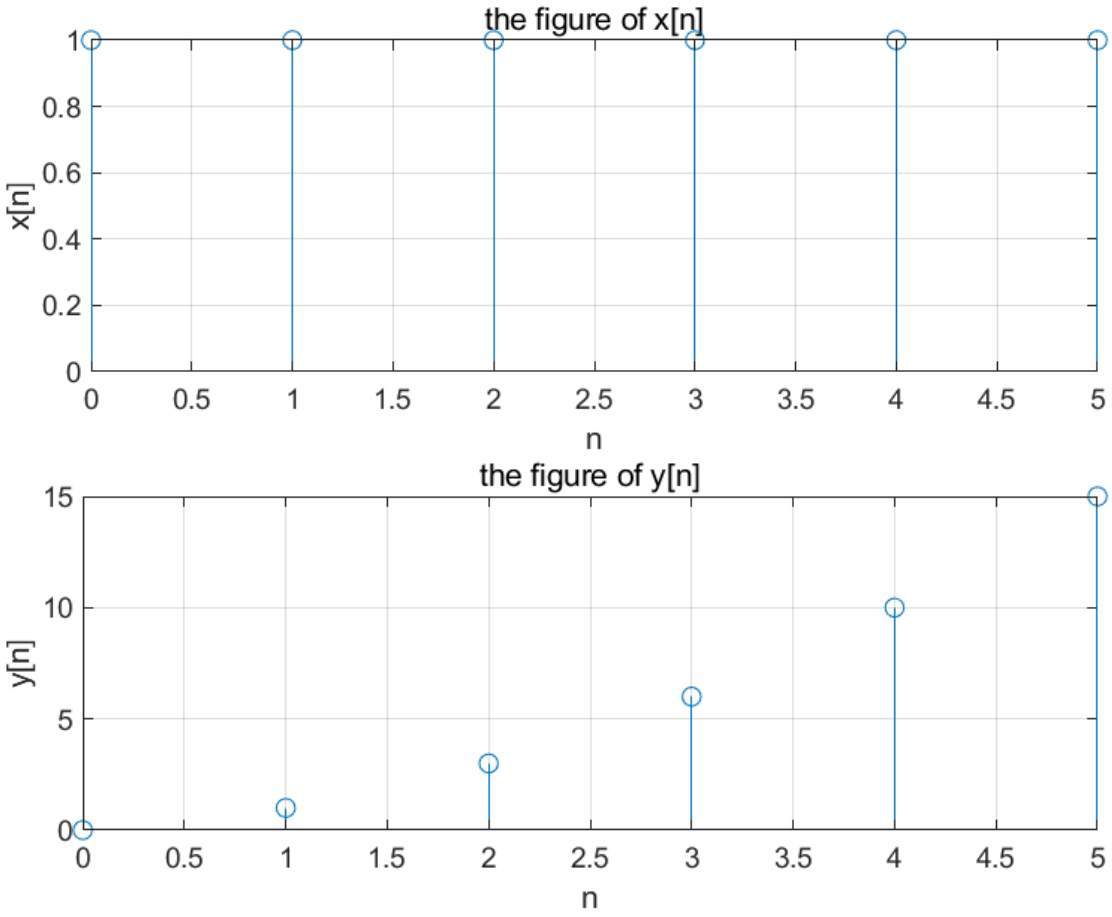
ef

Codes:

```
clc
clear
x=[ones(1,6)];
nx=[0:5];
b=[0:5];
a=1;
ny=[0:5];
y=filter(b,a,x);
figure(1);
subplot(2,1,1);
stem(nx,x);
xlabel('n');
```

```
ylabel('x[n]');
title('the figure of x[n]');
grid on;
subplot(2,1,2);
stem(ny,y);
xlabel('n');
ylabel('y[n]');
title('the figure of y[n]');
grid on;
```

figure:



Explanation

(e) Store the input signal x[n] and impulse response h[n] in the vectors x and h, respectively, over the interval 0≤n≤5

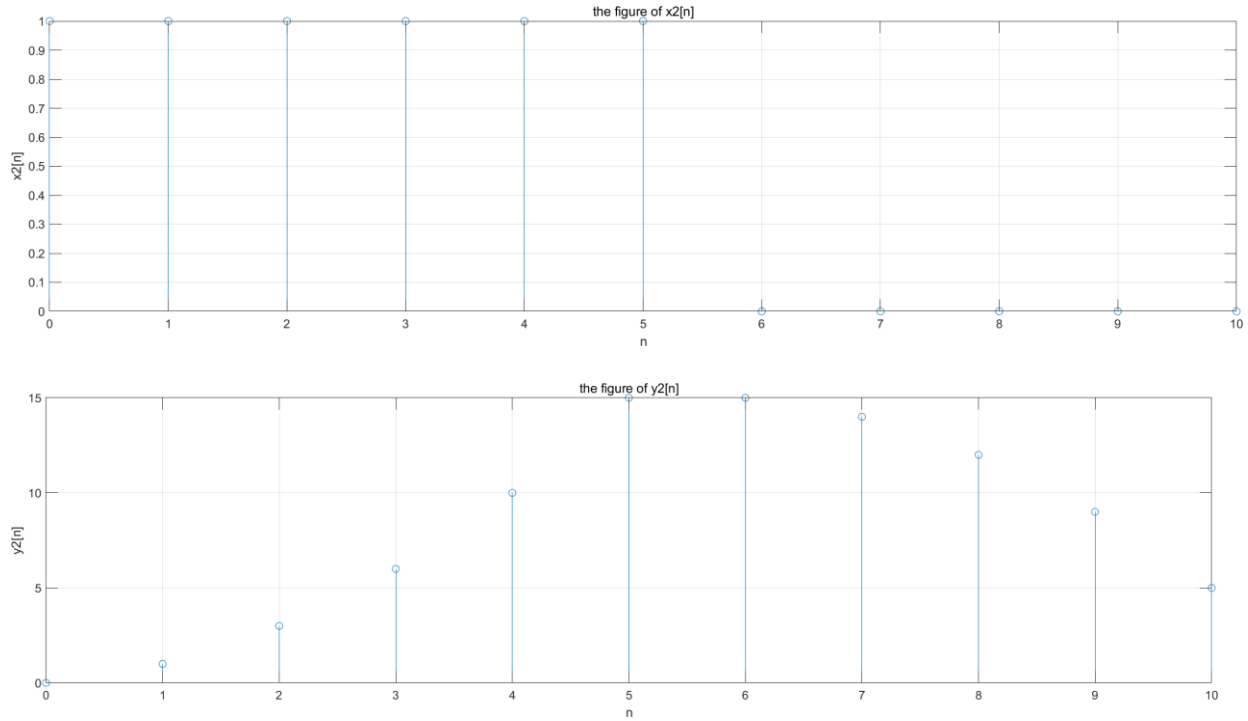
(f) To use the filter function to compute the output, map the impulse response h[n] to the coefficients of the difference equation, setting b = h and a = 1. Then, calculate the output y over the interval 0≤n≤5 using y = filter(h, 1, x), and plot the result with stem(ny, y).

codes:

```
clc
clear
```

```
x2=[ones(1,6) zeros(1,5)];
nx2=[0:10];
h=[0:5];
a=1;
y2=filter(h,a,x2);
figure(1);
subplot(2,1,1);
stem(nx2,x2);
xlabel('n');
ylabel('x2[n]');
title('the figure of x2[n]');
grid on;
subplot(2,1,2);
stem([0:10],y2);
xlabel('n');
ylabel('y2[n]');
title('the figure of y2[n]');
grid on;
```

Figure



Explanations

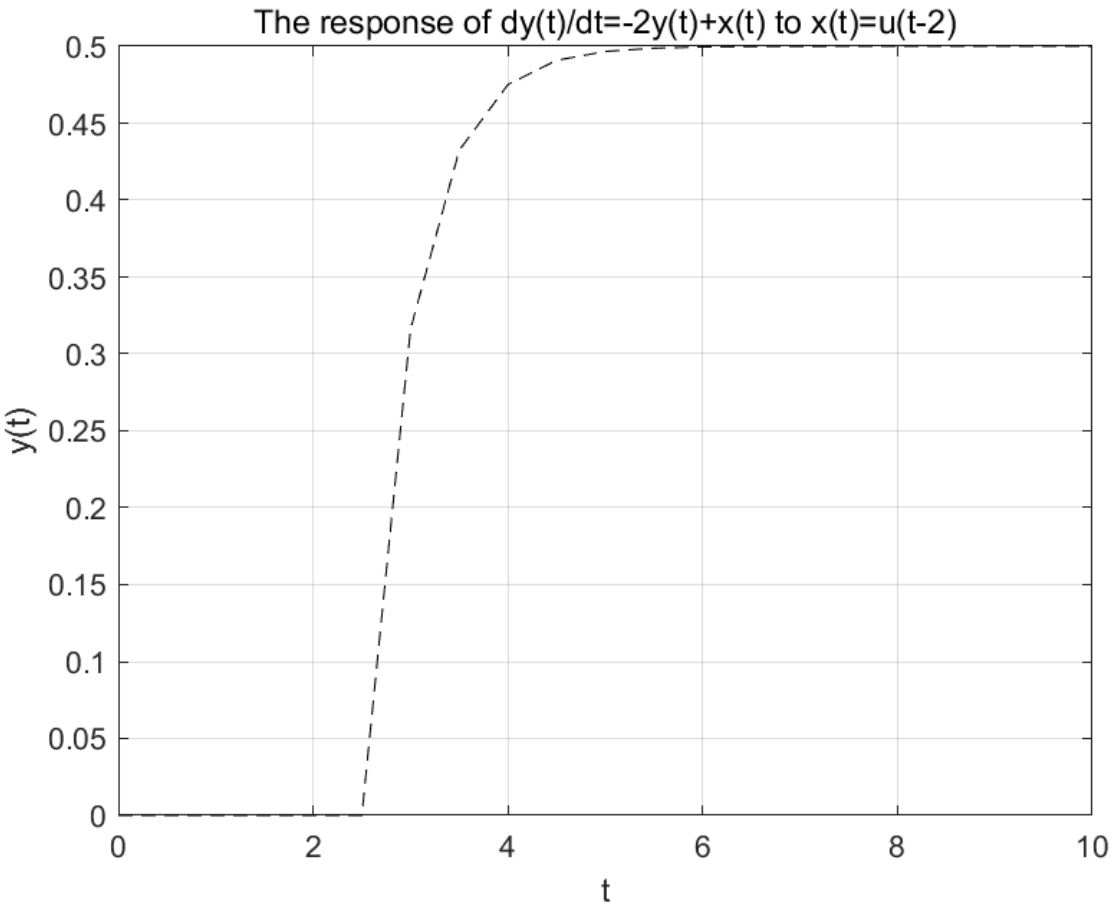
The image generated by filter is exactly the same as the figure in 2.2,which proves y=x[n]*h[n]=filter(h,1,x).

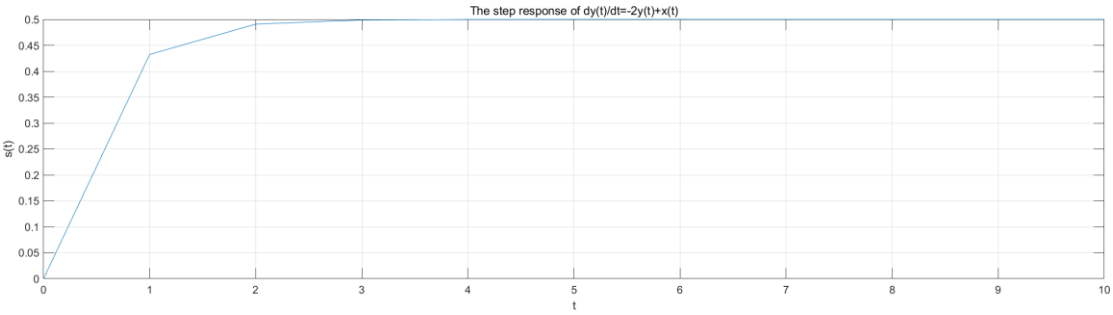
2.3

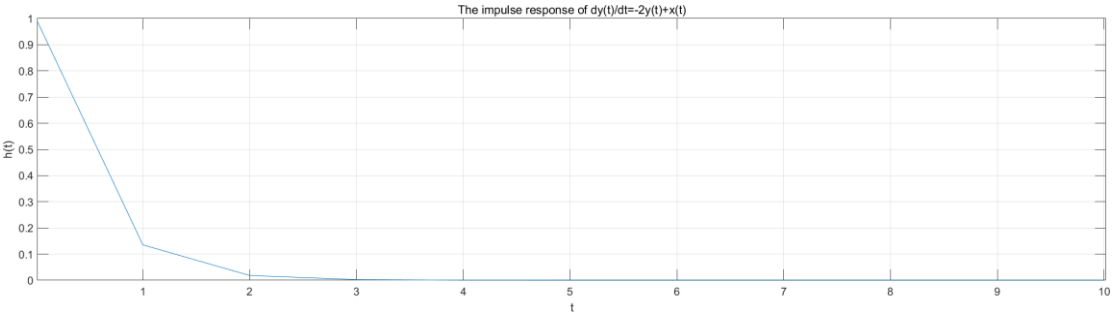
a

Codes:

```
clc
clear
t=[0:0.5:10];
x=Heaviside(t-2);
b=1;
a=[1 2];
y=lsim(b,a,x,t);
figure(1);
plot(t,y,'k--');
xlabel('t');
ylabel('y(t)');
title('The response of dy(t)/dt=-2y(t)+x(t) to x(t)=u(t-2)');
grid on;
t1=[0:1:10];
b1=1;
a1=[1 2];
s=step(b1,a1,t1);
h=impulse(b1,a1,t1);
figure(2);
subplot(2,1,1);
plot(t1,s);
xlabel('t');
ylabel('s(t)');
title('The step response of dy(t)/dt=-2y(t)+x(t)');
grid on;
subplot(2,1,2);
plot(t1,h);
xlabel('t');
ylabel('h(t)');
title('The impulse response of dy(t)/dt=-2y(t)+x(t)');
grid on;
figure
```







b

Codes:

```
clc;
clear;
```

```
% Define the time vector
t = 0:1:10;

% Define the transfer function parameters
b = 1;
a = [1 0.5];

% Step and impulse responses
s = step(b, a, t);
h = impulse(b, a, t);

% Define symbolic variable for t
syms t_sym;
h2_sym = 2 * (1 - exp(-t_sym/2)) * heaviside(t_sym);

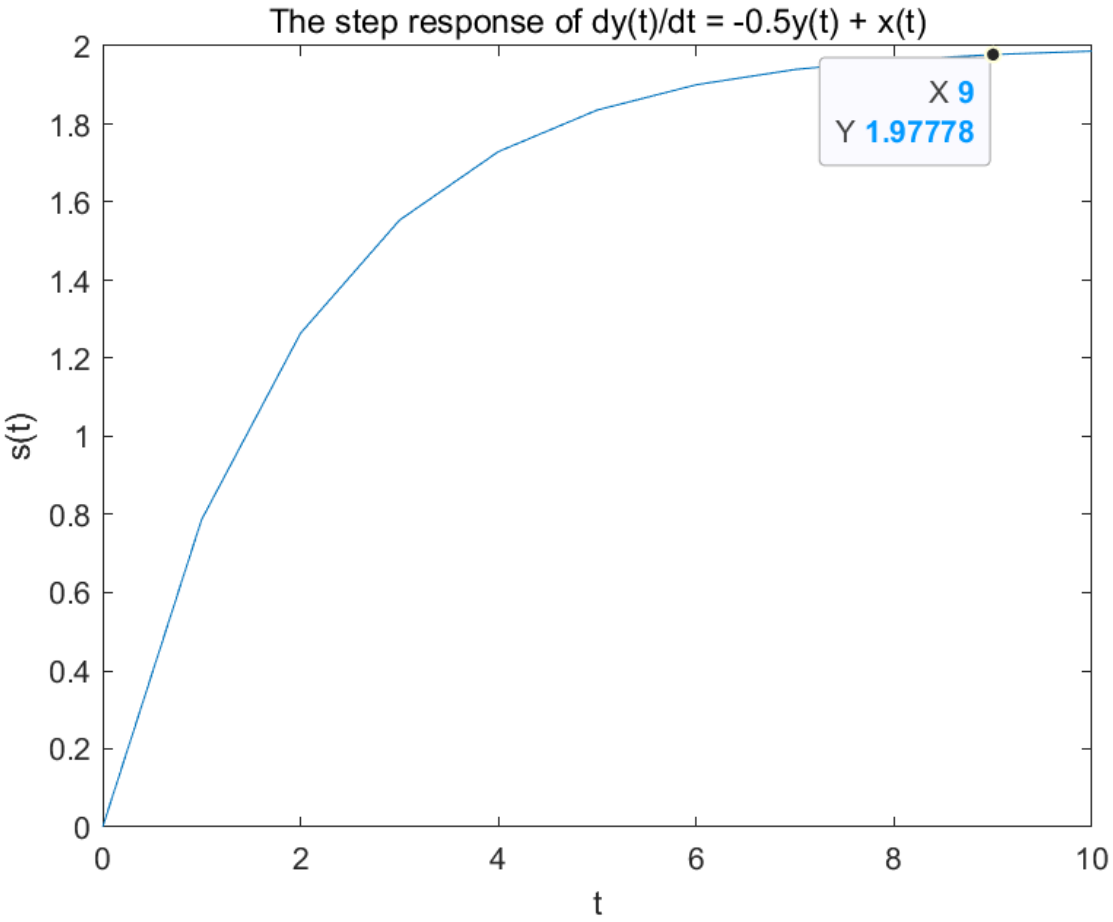
% Derivative of the symbolic function
h2_deriv_sym = diff(h2_sym, t_sym);

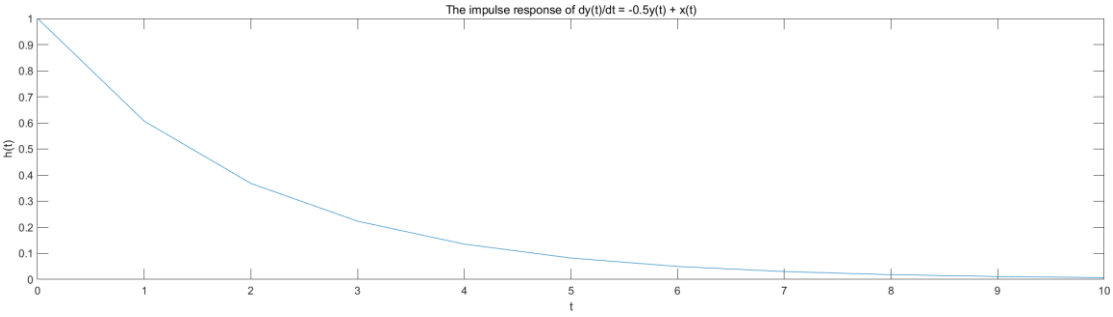
% Plot the step response
figure(1);
plot(t, s);
xlabel('t');
ylabel('s(t)');
title('The step response of dy(t)/dt = -0.5y(t) + x(t)');

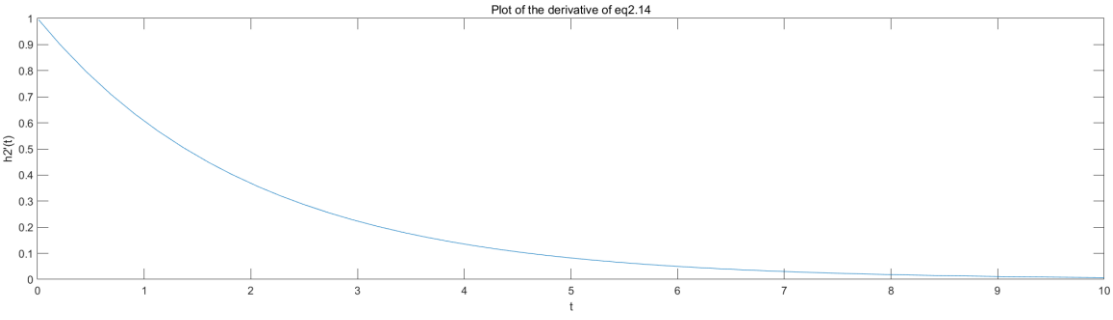
% Plot the impulse response and the derivative of eq2.14
figure(2);
subplot(2, 1, 1);
plot(t, h);
xlabel('t');
ylabel('h(t)');
title('The impulse response of dy(t)/dt = -0.5y(t) + x(t)');

% Plot the symbolic derivative
subplot(2, 1, 2);
fplot(h2_deriv_sym, [0 10]);
xlabel('t');
ylabel('h2''(t)');
title('Plot of the derivative of eq2.14');
```

figure







Explanation:

1. **Step Response Comparison**:
    - The first plot generated shows the step response s(t) of the causal LTI system given by dy(t)/dt=−0.5y(t)+x(t)
    - Comparing this plot with Figure 2.6, we observe that the computed step

response closely matches the shape and asymptotic behavior shown in Figure 2.6, where the response initially rises and gradually approaches a steady-state value around 2 as t→10.
    - This consistency suggests that the system's step response has been correctly computed using MATLAB's step function, matching the expected theoretical behavior shown in Figure 2.6.
2. **Impulse Response Comparison**:
    - The second plot displays the impulse response h(t) obtained using MATLAB's impulse function. It starts at 1 and decays exponentially towards zero, which aligns with the behavior expected from an impulse response.
    - The third plot, representing the exact impulse response (derived as the derivative of the step response s(t), also exhibits an exponential decay pattern, closely matching the computed impulse response in the second plot.
    - This similarity confirms that the response obtained using impulse accurately represents the system's theoretical impulse response as defined by the derivative of s(t) in Eq. (2.14).

In summary, the computed step response and impulse response align well with the theoretical expectations and Figure 2.6, validating the correctness of the MATLAB simulations.

2.4

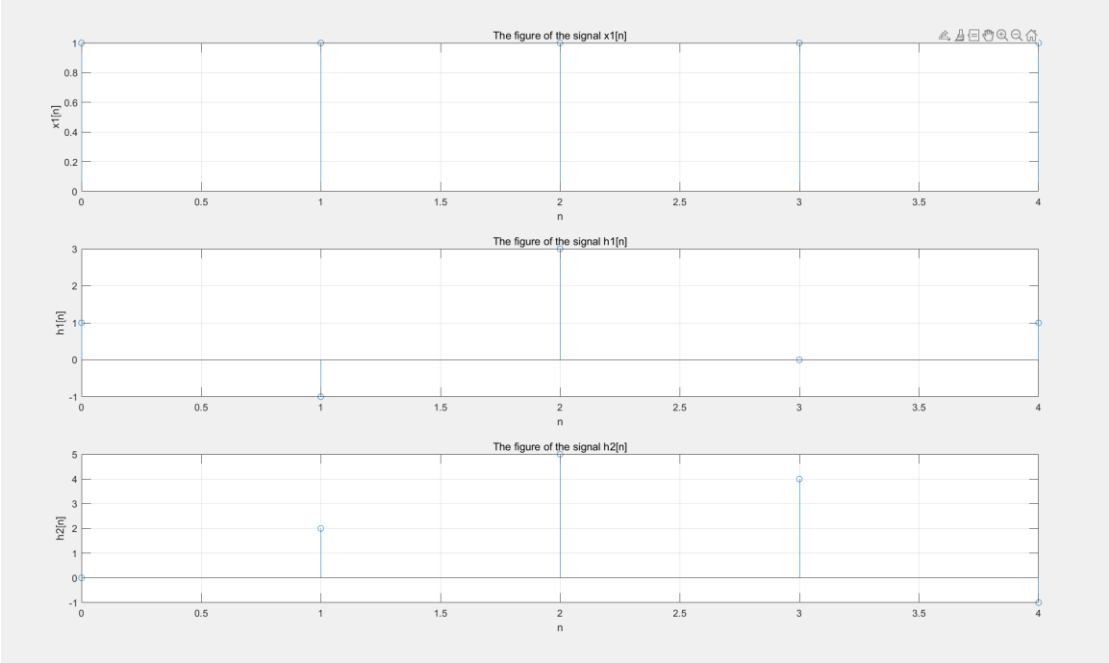
A

Codes:

```
clc
clear
nx1=[0:4];
x1=[1 1 1 1 1];
h1=[1 -1 3 0 1];
h2=[0 2 5 4 -1];
subplot(3,1,1);
stem(nx1,x1);
xlabel('n');
ylabel('x1[n]');
title('The figure of the signal x1[n]');
grid on;
subplot(3,1,2);
stem(nx1,h1);
xlabel('n');
ylabel('h1[n]');
title('The figure of the signal h1[n]');
```

```
grid on;
subplot(3,1,3);
stem(nx1,h2);
xlabel('n');
ylabel('h2[n]');
title('The figure of the signal h2[n]');
grid on;
```

figure:

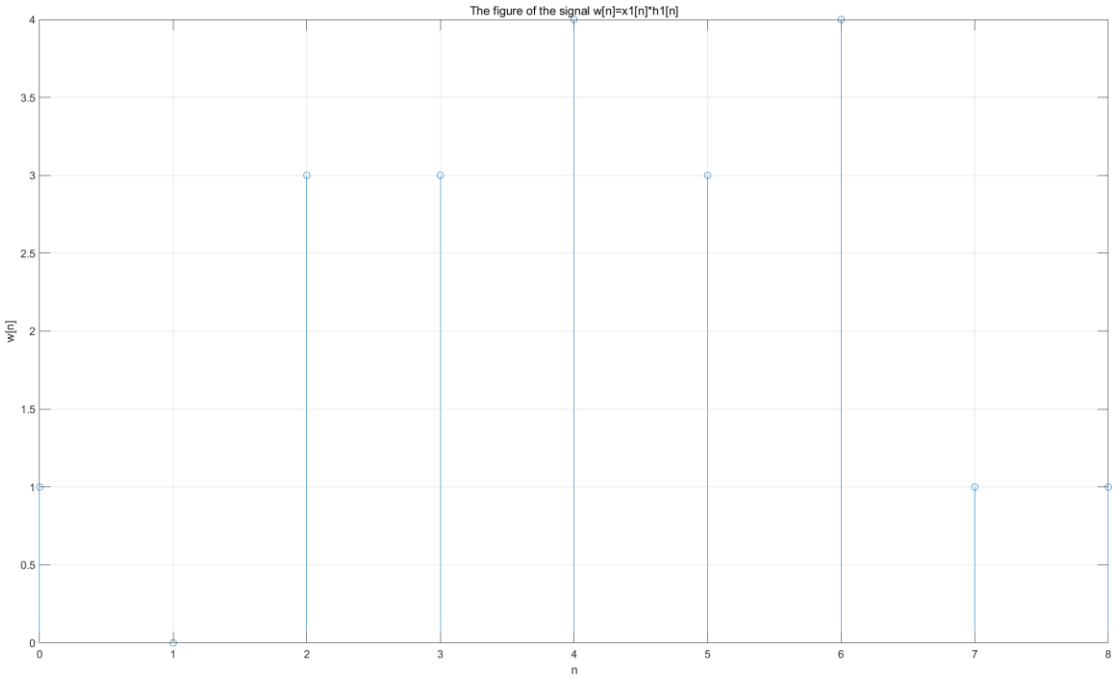


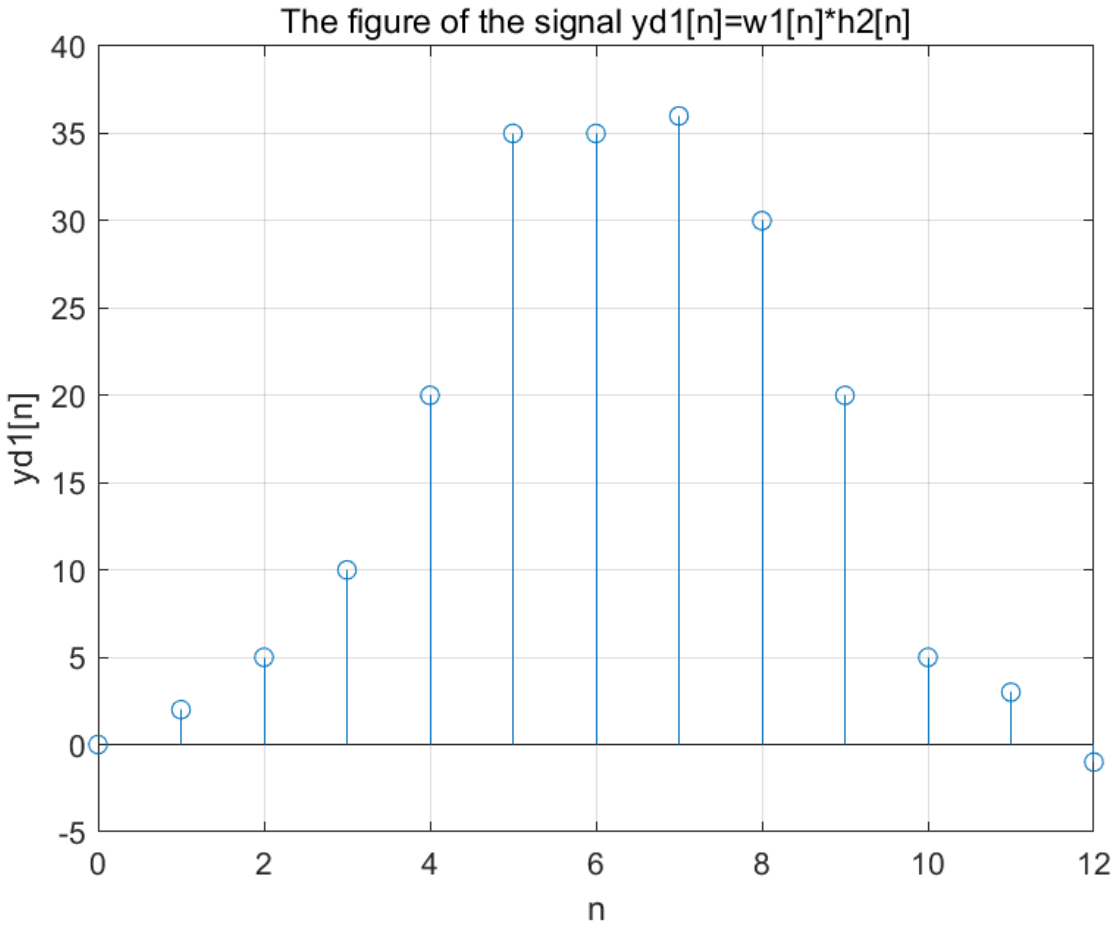
d

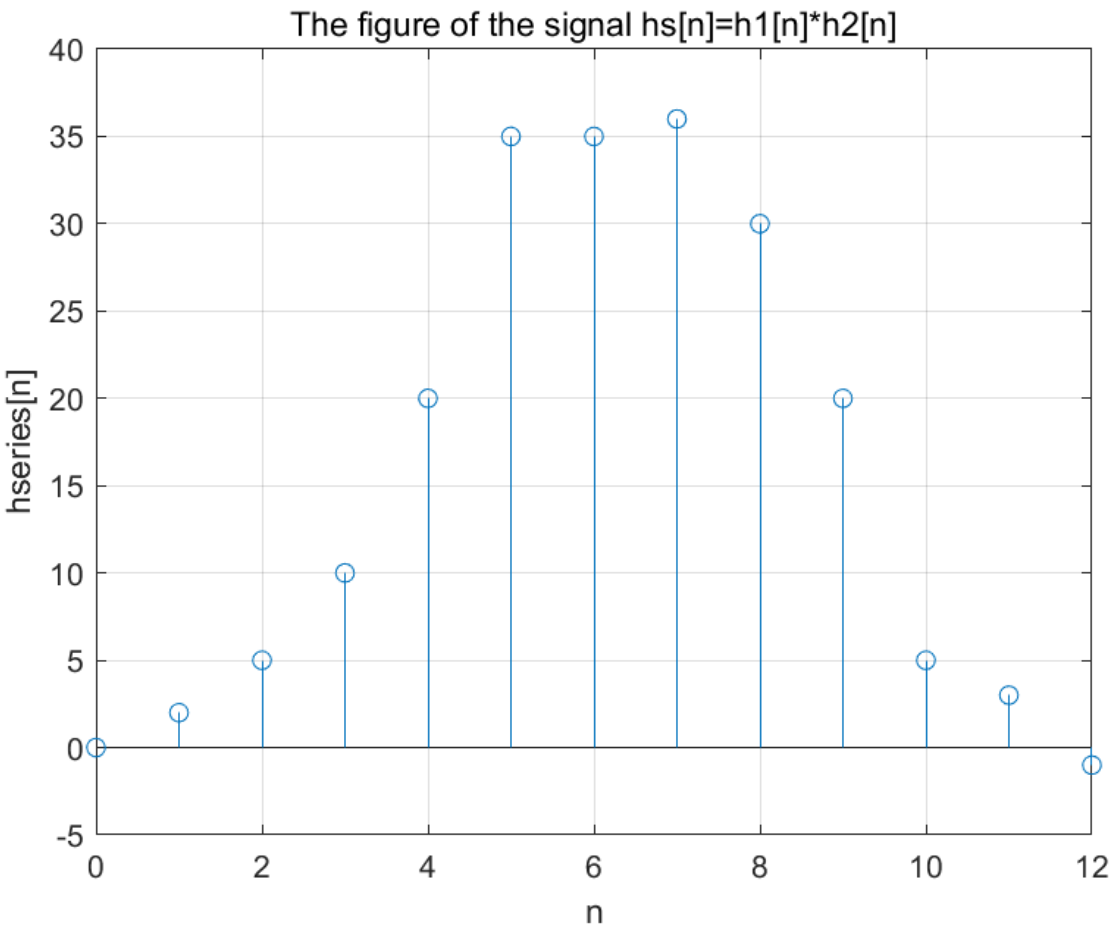
codes:

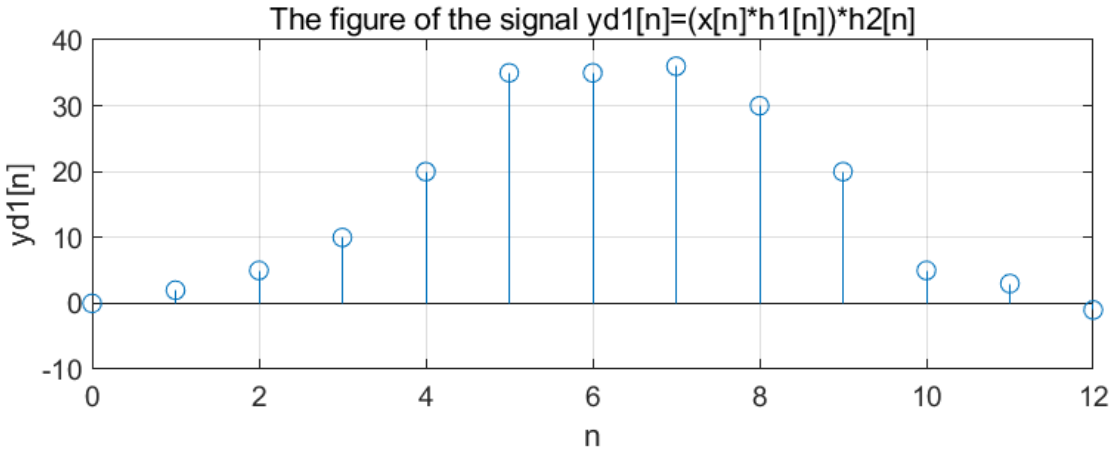
```
clc
clear
nx1=[0:4];
x1=[1 1 1 1 1];
h1=[1 -1 3 0 1];
h2=[0 2 5 4 -1];
nw=[0:8];
w=conv(x1,h1);
nw=[0:length(w)-1];
figure(1);
stem(nw,w);
xlabel('n');
ylabel('w[n]');
title('The figure of the signal w[n]=x1[n]*h1[n]');
grid on;
```

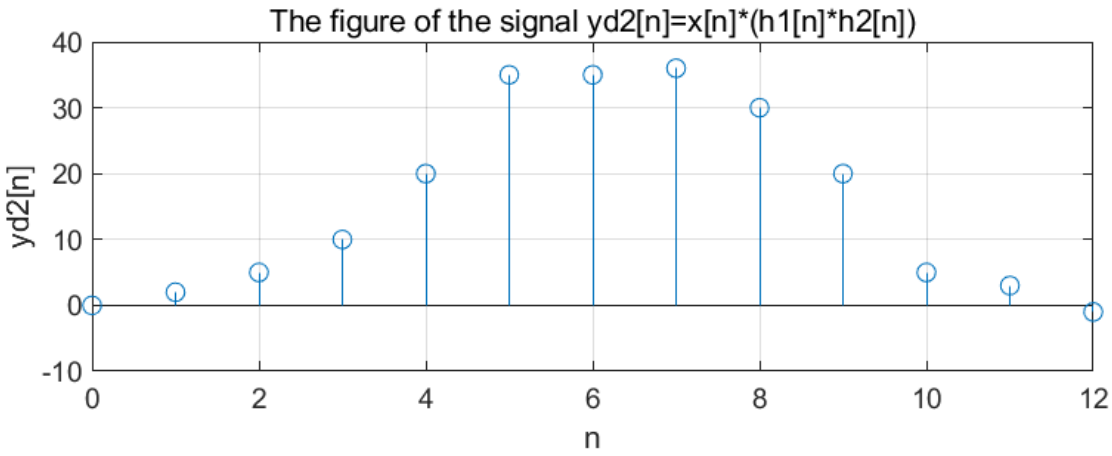
```
yd1=conv(w,h2);
nyd1=[0:length(yd1)-1];
figure(2);
stem(nyd1,yd1);
xlabel('n');
ylabel('yd1[n]');
title('The figure of the signal yd1[n]=w1[n]*h2[n]');
grid on;
hs=conv(h1,h2);
nhs=[0:length(hs)-1];
figure(3);
stem(nyd1,yd1);
xlabel('n');
ylabel('hseries[n]');
title('The figure of the signal hs[n]=h1[n]*h2[n]');
grid on;
yd2=conv(x1,hs);
nyd2=[0:length(yd2)-1];
figure(4);
subplot(2,1,1);
stem(nyd1,yd1);
xlabel('n');
ylabel('yd1[n]');
title('The figure of the signal yd1[n]=(x[n]*h1[n])*h2[n]');
grid on;
subplot(2,1,2);
stem(nyd2,yd2);
xlabel('n');
ylabel('yd2[n]');
title('The figure of the signal yd2[n]=x[n]*(h1[n]*h2[n])');
grid on;
figure
```


The figure of the signal w[n]=x1[n]*h1[n]
w[n]
n


The figure of the signal yd1[n]=w1[n]*h2[n]
yd1[n]
n


The figure of the signal hs[n]=h1[n]*h2[n]
hseries[n]
n
40
35
30
25
20
15
10
5
0
-5
0
2
4
6
8
10
12


The figure of the signal yd1[n]=(x[n]*h1[n])*h2[n]
yd1[n]
n
40
30
20
10
0
-10
0
2
4
6
8
10
12



Explanation:

The figures of yd1[n] and yd2[n] are exactly the same,which proves the associative property (x[n]*h1[n])*h2[n]=x[n]*(h1[n]*h2[n])

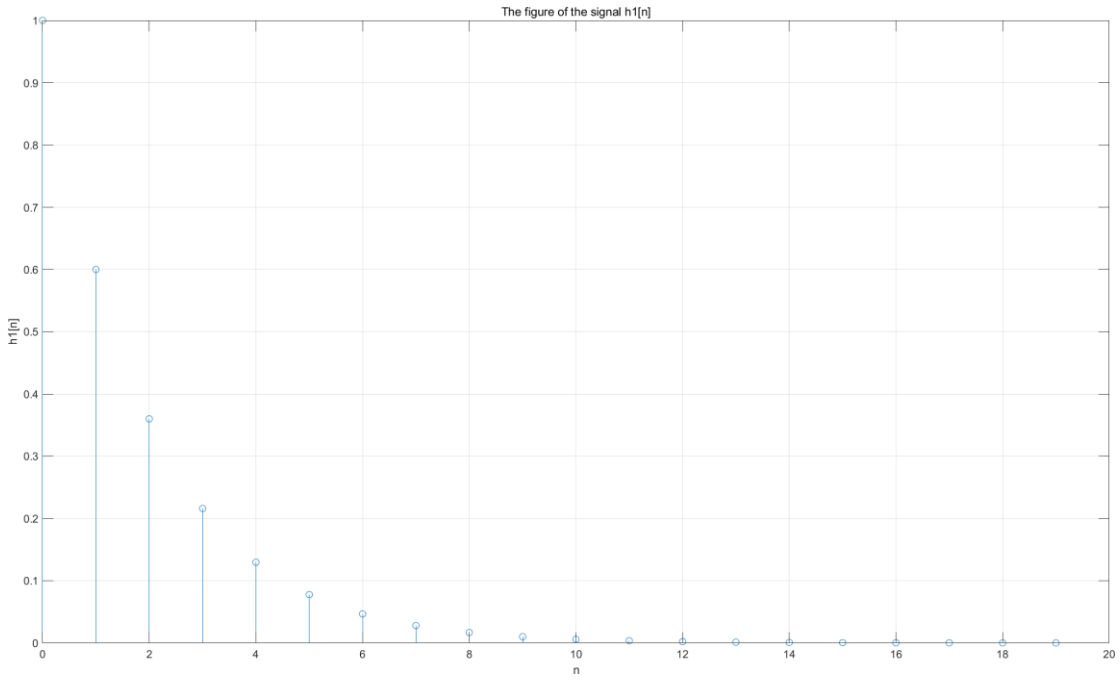
2.5

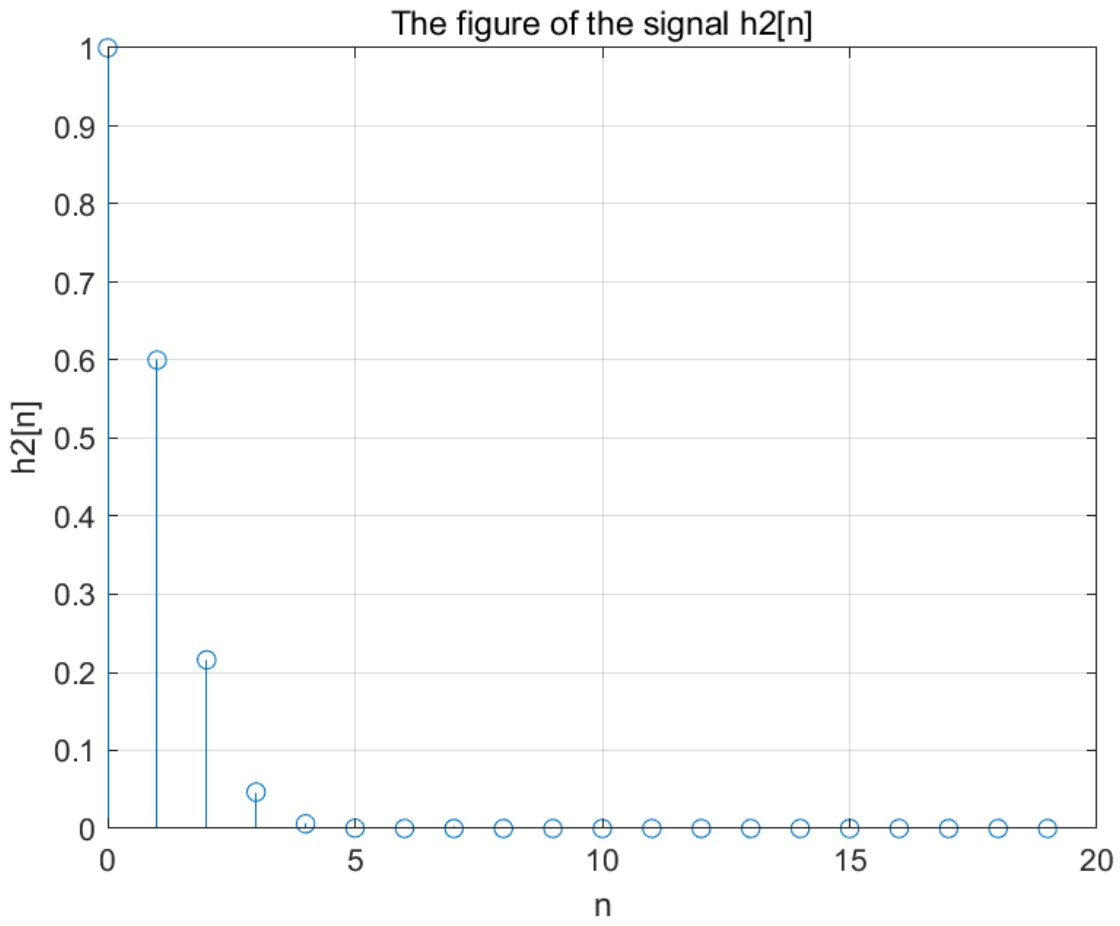
d

```
clc
clear
a1=[1 -3/5];
b1=[1];
n=[0:19];
x1=[1 zeros(1,19)];
h1=filter(b1,a1,x1);
figure(1);
stem(n,h1);
xlabel('n');
ylabel('h1[n]');
title('The figure of the signal h1[n]');
grid on;
y2(1)=1;
for k=2:20
y2(k)=((3/5)^(k-1))*y2(k-1)+x1(k);
end
h2=[];
for m=0:19
h2(end+1)=y2(m+1);
end
figure;
stem(n,h2);
xlabel('n');
ylabel('h2[n]');
title('The figure of the signal h2[n]');
```

grid on;

figure:





efg)code:

```
clc
clear all
%.....(e).....%
a1=[1 -3/5];
b1=[1];
n=[0:19];
x1=[ones(1,20)];
s1=filter(b1,a1,x1);
figure(1);
subplot(2,1,1);
stem(n,s1);
xlabel('n');
ylabel('s1[n]');
title('The figure of the signal s1[n]');
grid on;

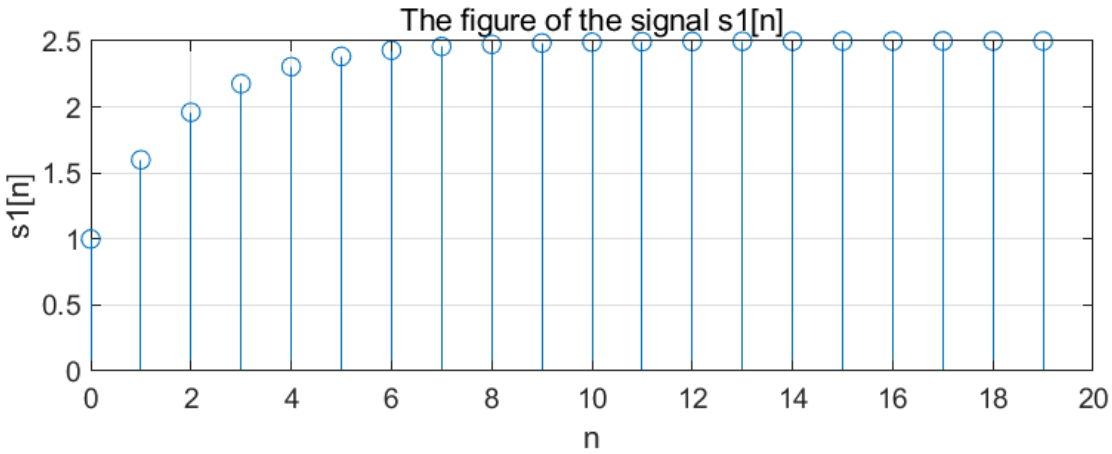

y2(1)=1;
for k=2:20
y2(k)=((3/5).^(k-1))*y2(k-1)+x1(k);
end
s2=[];
for m=0:19
s2(end+1)=y2(m+1);
end
subplot(2,1,2);
stem(n,s2);
xlabel('n');
ylabel('s2[n]');
title('The figure of the signal s2[n]');
grid on;
%.....(f).....%

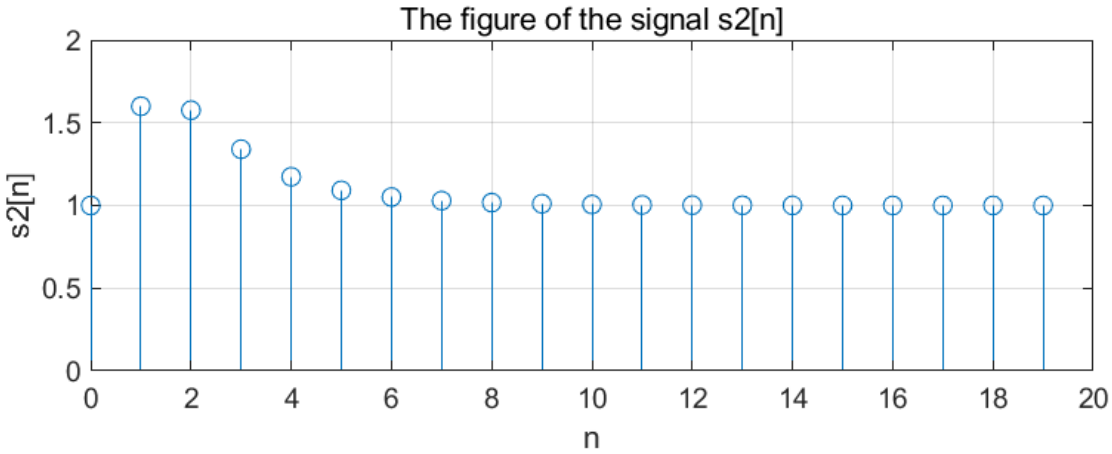
x0=[1 zeros(1,19)];
h1=filter(b1,a1,x0);
y0(1)=1;
for k=2:20
y0(k)=((3/5)^(k-1))*y0(k-1)+x0(k);
end
h2=[];
for m=0:19
h2(end+1)=y0(m+1);
end
z1=conv(h1,x1);
```

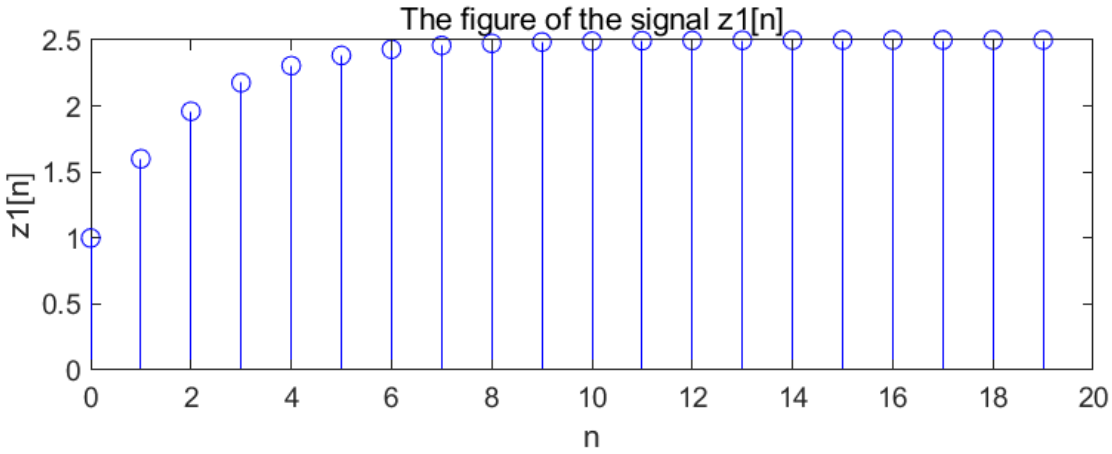
```
Z1=[];
for i1=1:20
Z1(end+1)=z1(i1);
end
figure(2);
subplot(2,1,1);
stem(n,Z1,'b');
xlabel('n');
ylabel('z1[n]');
title('The figure of the signal z1[n]');

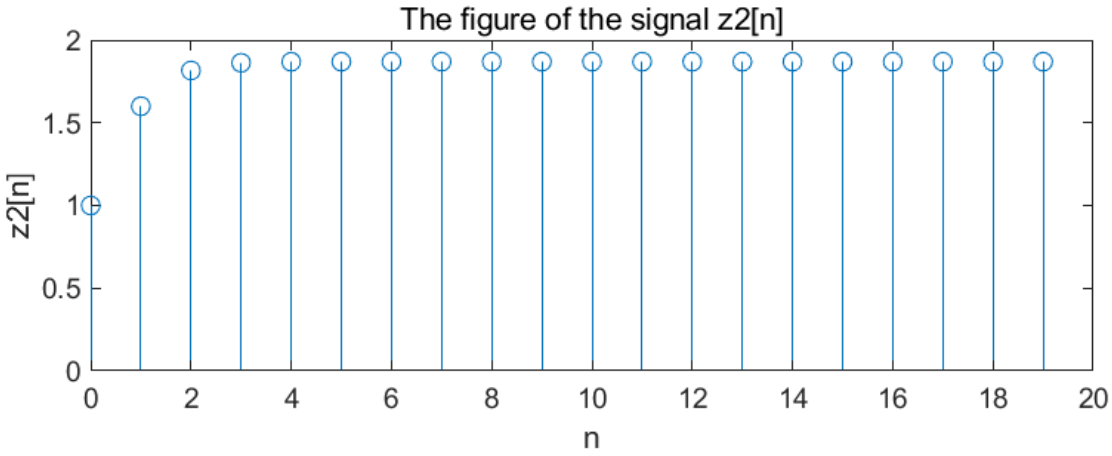
z2=conv(h2,x1);
Z2=[];
for i2=1:20
Z2(end+1)=z2(i2);
end
subplot(2,1,2);
stem(n,Z2);
xlabel('n');
ylabel('z2[n]');
title('The figure of the signal z2[n]');
%.....(g).....%
figure(3);
stem(n,s1,'b');
hold on;
stem(n,Z1,'k');
xlabel('n');
ylabel('s1[n]&z1[n]');
title('The figure of the signals s1[n](blue) and z1[n](black)');
grid on;
figure(4);
subplot(2,1,1);
stem(n,s2);
xlabel('n');
ylabel('s2[n]');
title('The figure of the signal s2[n]');
grid on;
subplot(2,1,2);
stem(n,Z2);
xlabel('n');
ylabel('z2[n]');
title('The figure of the signal z2[n]');
grid on;
```

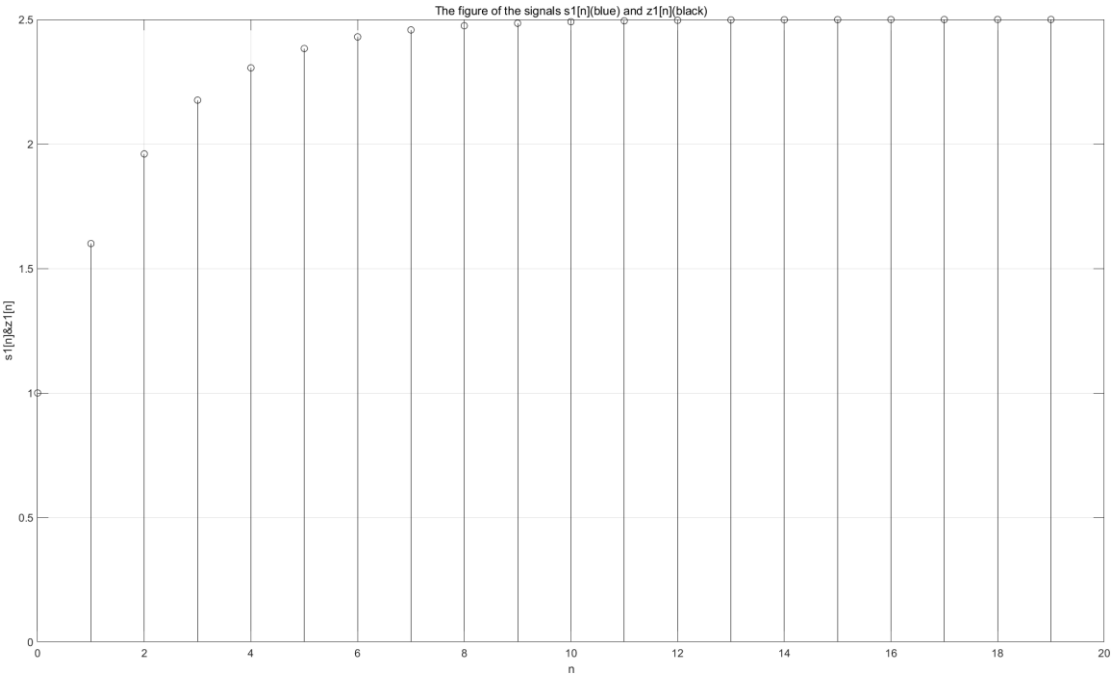
figure

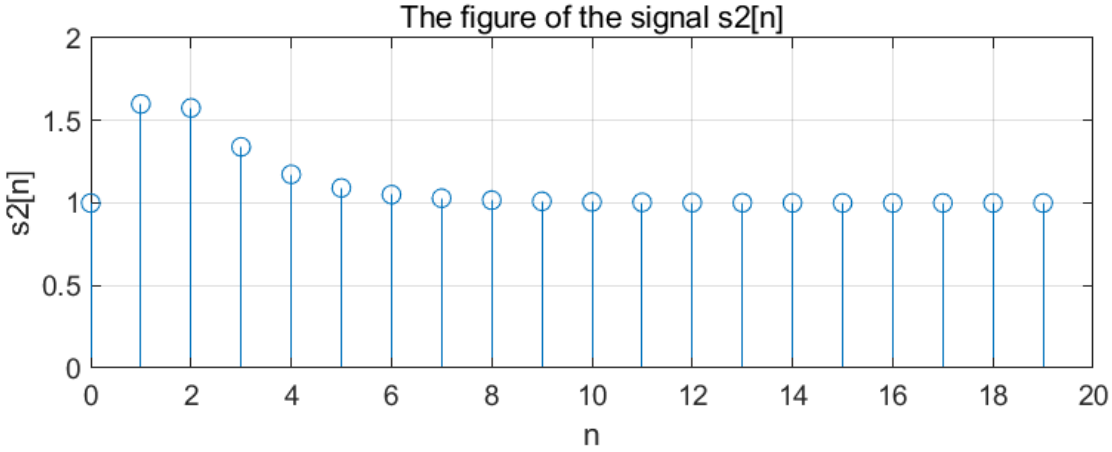

The figure of the signal s1[n]
s1[n]
n

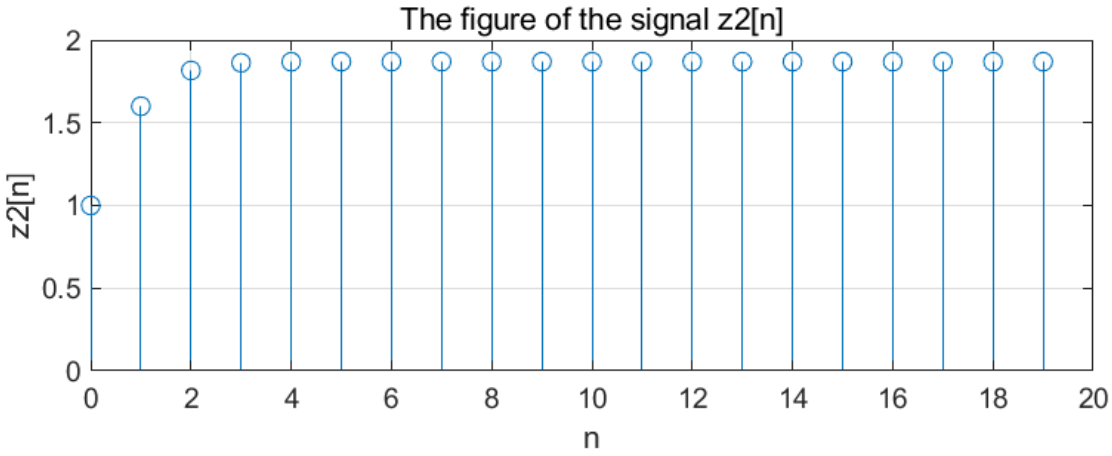

The figure of the signal s2[n]
s2[n]
n


The figure of the signal z1[n]
z1[n]
n


The figure of the signal z2[n]
z2[n]
n


The figure of the signals s1[n](blue) and z1[n](black)
s1[n]&z1[n]
n


The figure of the signal s2[n]
s2[n]
n



Explanation:

s[n] is calculated directly using a for loop, which allows each output to adapt to changes in the system's behavior over time, accurately capturing the time-varying response.
z[n] is calculated by convolving h2[n] with u[n].

- In System 1 (time-invariant), s1[n] and z1[n] are nearly identical because convolution can accurately represent the system's response, given that the system's characteristics do not change over time.
- In System 2 (time-varying), s2[n] and z2[n] differ because z2[n] (computed via convolution) assumes time invariance and therefore fails to capture the system's time-varying nature. In contrast, s2[n], calculated iteratively, reflects the system's changing behavior more accurately.

2.7

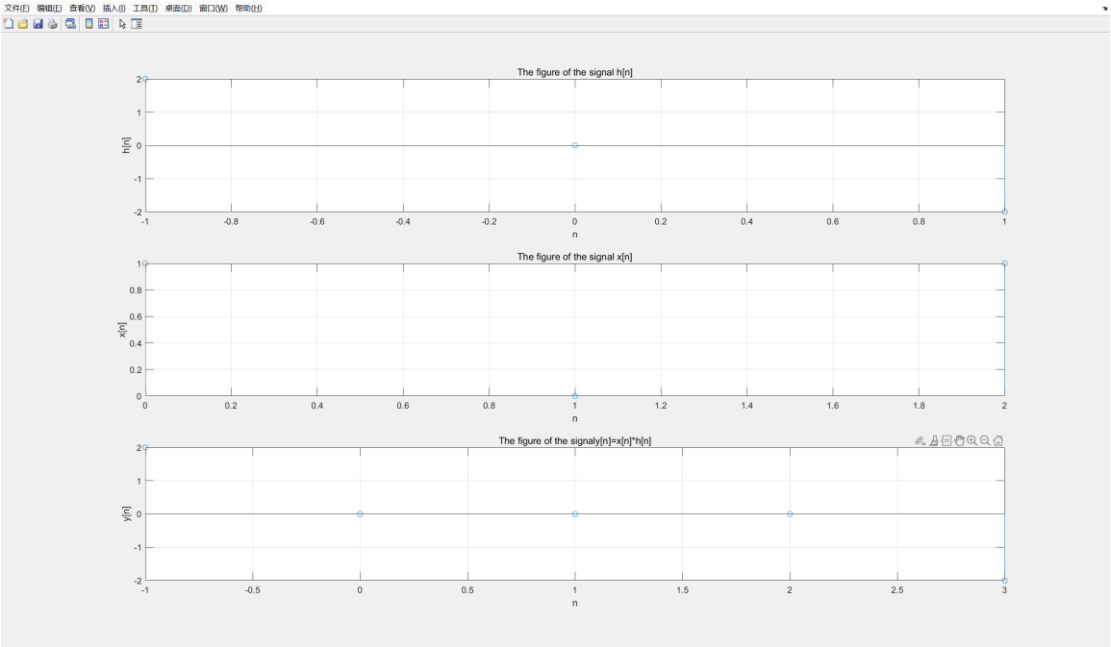
A

Codes:

```
clc
clear
nh=[-1 0 1];
h=[2 0 -2];
nx=[0 1 2];
x=[1 0 1];
k1=nh(1)+nx(1);
k2=length(h)+length(x)-2;
ny=[k1:k1+k2];
y=conv(h,x);
subplot(3,1,1);
stem(nh,h);
xlabel('n');
ylabel('h[n]');
title('The figure of the signal h[n]');
```

```
grid on;
subplot(3,1,2);
stem(nx,x);
xlabel('n');
ylabel('x[n]');
title('The figure of the signal x[n]');
grid on;
subplot(3,1,3);
stem(ny,y);
xlabel('n');
ylabel('y[n]');
title('The figure of the signaly[n]=x[n]*h[n]');
grid on;
```

figure:



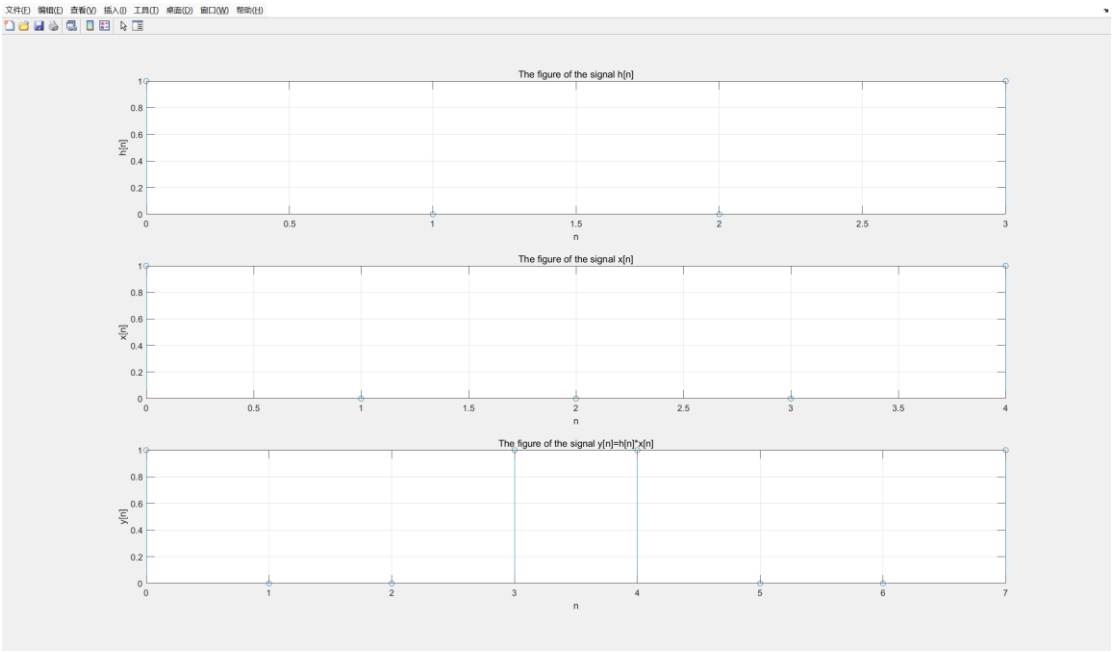
B

Codes:

```
clc
clear
N=4;
M=5;
nx=[0:4];
nh=[0:3];
h=[1 0 0 1];
x=[1 0 0 0 1];
subplot(3,1,1);
```

```
stem(nh,h);
xlabel('n');
ylabel('h[n]');
title('The figure of the signal h[n]');
grid on;
subplot(3,1,2);
stem(nx,x);
xlabel('n');
ylabel('x[n]');
title('The figure of the signal x[n]');
grid on;
y=conv(h,x);
ny=[0:length(y)-1];
subplot(3,1,3);
stem(ny,y);
xlabel('n');
ylabel('y[n]');
title('The figure of the signal y[n]=h[n]*x[n]');
grid on;
```

figure:



Expanation

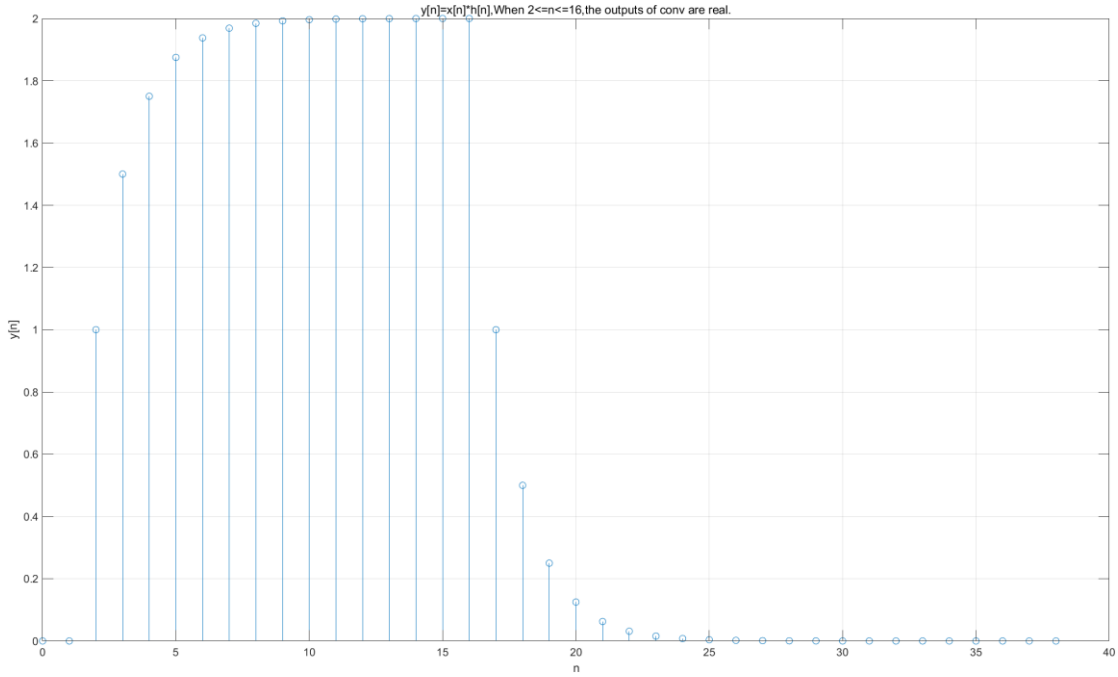
According to the figures, Length of y=nymax-nymin+1=(7-0)+1=length of x +length of h-1=[(4-0)+1]+ [(3-0)+1]-1=8=M+N-1

C:

Codes:

```
clc
clear
nx1=[0:24];
nh=[0:14];
nx2=[2:24];
x=[zeros(1,2) (1/2).^(nx2-2)];
h=[ones(1,15)];
y=conv(h,x);
ny=[0:length(y)-1];
stem(ny,y);
title('y[n]=x[n]*h[n],When 2<=n<=16,the outputs of conv are real.');
xlabel('n');
ylabel('y[n]');
grid on;
```

figure:



Explanation

y[n]=$\sum_{-\infty}^{\infty} x[k] \times h[n-k]$,

If n<2, when k<2,x[k]=0,therefore y[n] is unreal.

If n>16,when k<2,n-k>14,h[n-k]is unreal , therefore y[n] is unreal.

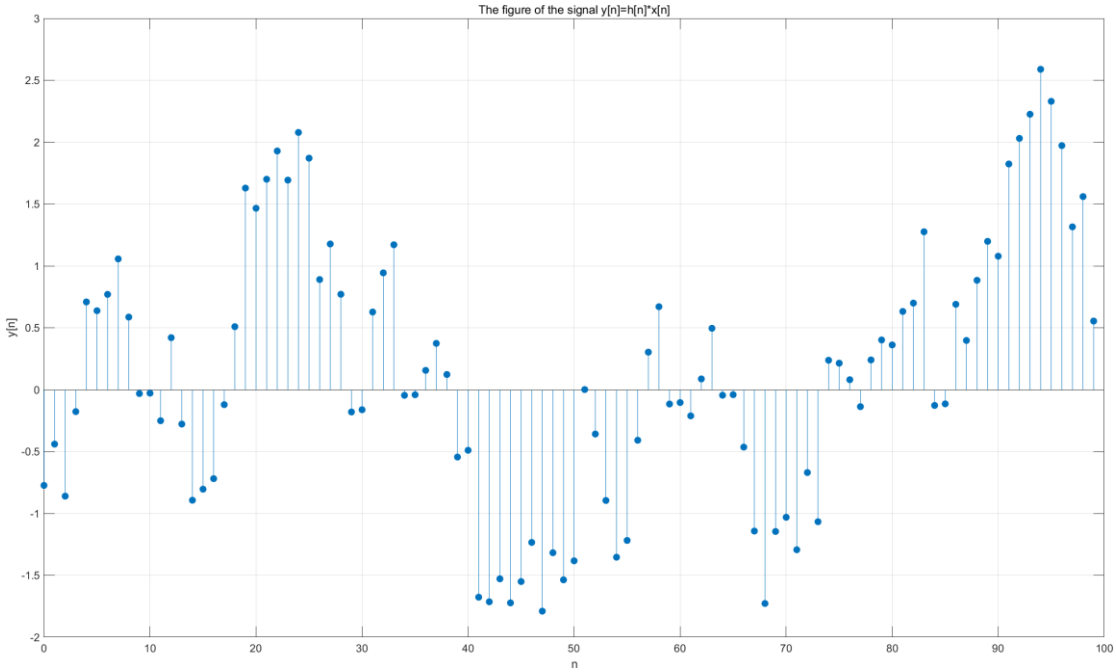
Thus if $2 \leq n \leq 16, when\ 2 \leq k \leq 16$, x[k]= $\frac{1}{2}^{k-2}$, $h[n-k]=1$, y[n]= $\sum_{-\infty}^{\infty} x[k] \times h[n-k], so\ it\ is\ real$

d)

codes:

```
clc
clear
nh=[0:9];
nx=[-9:99];
h=[(0.9).^nh];
x=[cos(nx.^2).*sin(2*pi*nx/5)];
y0=conv(h,x);
ny=[0:99];
y=[];
for k=0:99
y(end+1)=y0(k+10);
end
stem(ny,y,'fill');
xlabel('n');
ylabel('y[n]');
title('The figure of the signal y[n]=h[n]*x[n]');
grid on;
```

figure:



e)

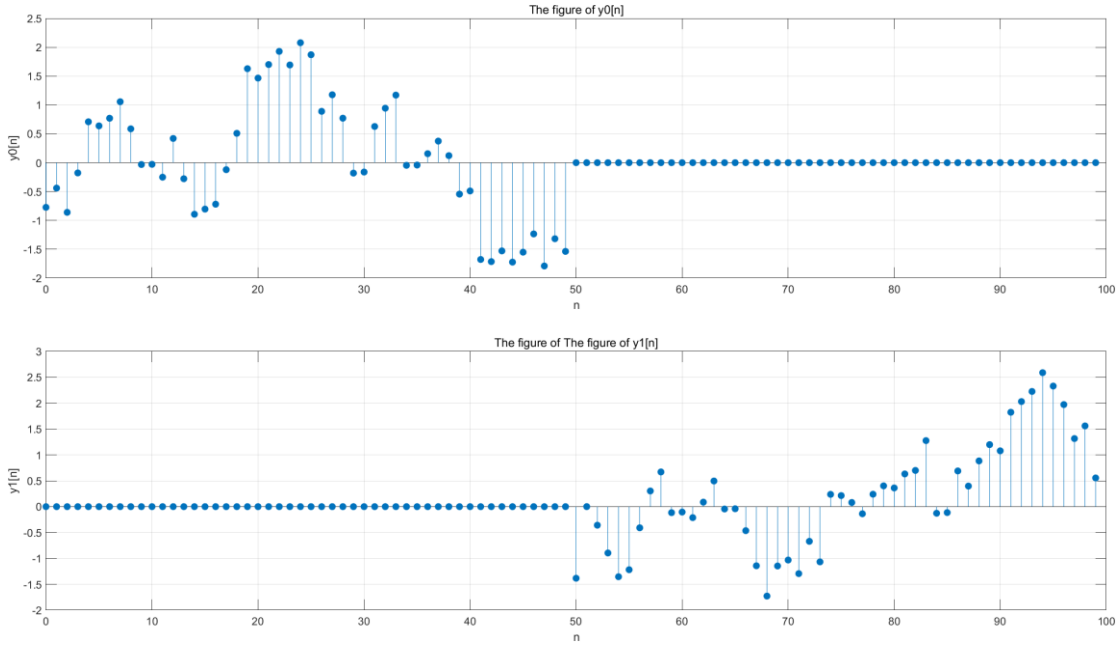
codes:

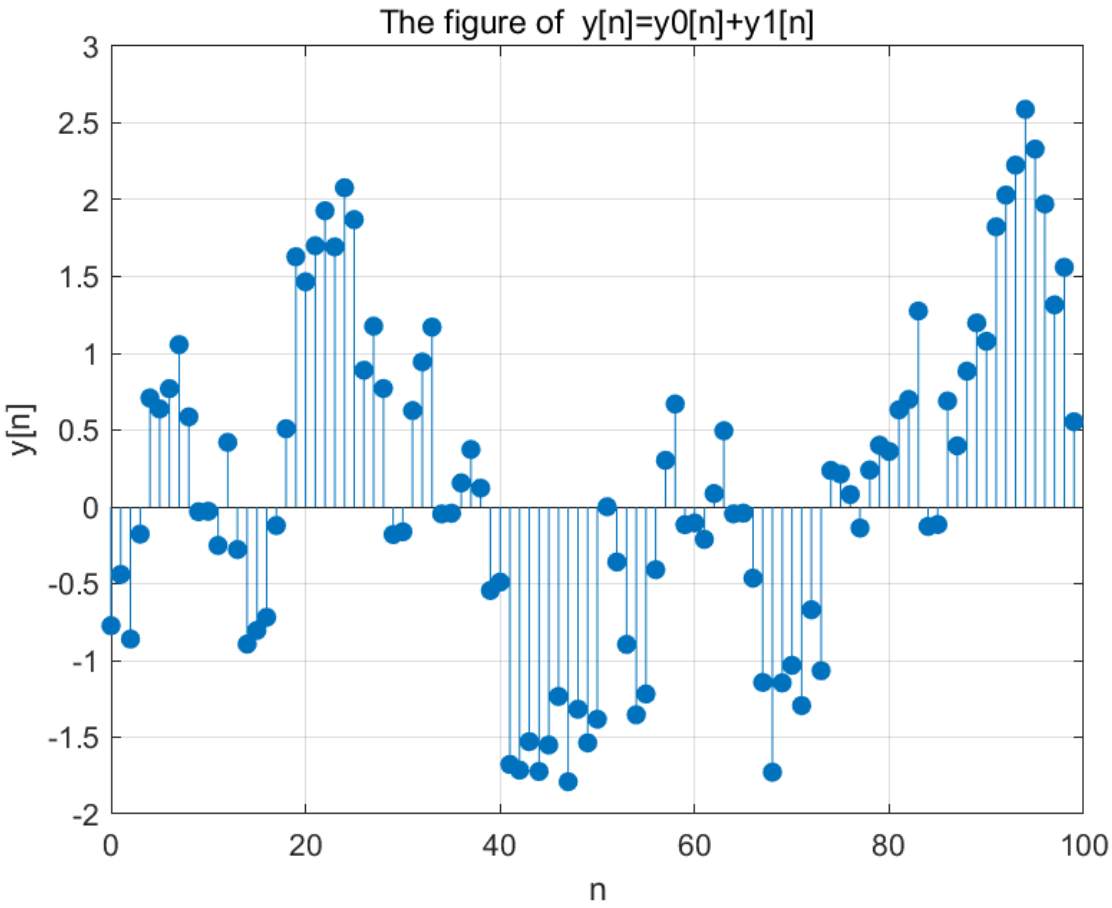
```
clc
clear
nh=[0:9];
nx0=[0:49];
nx1=[50:99];
```

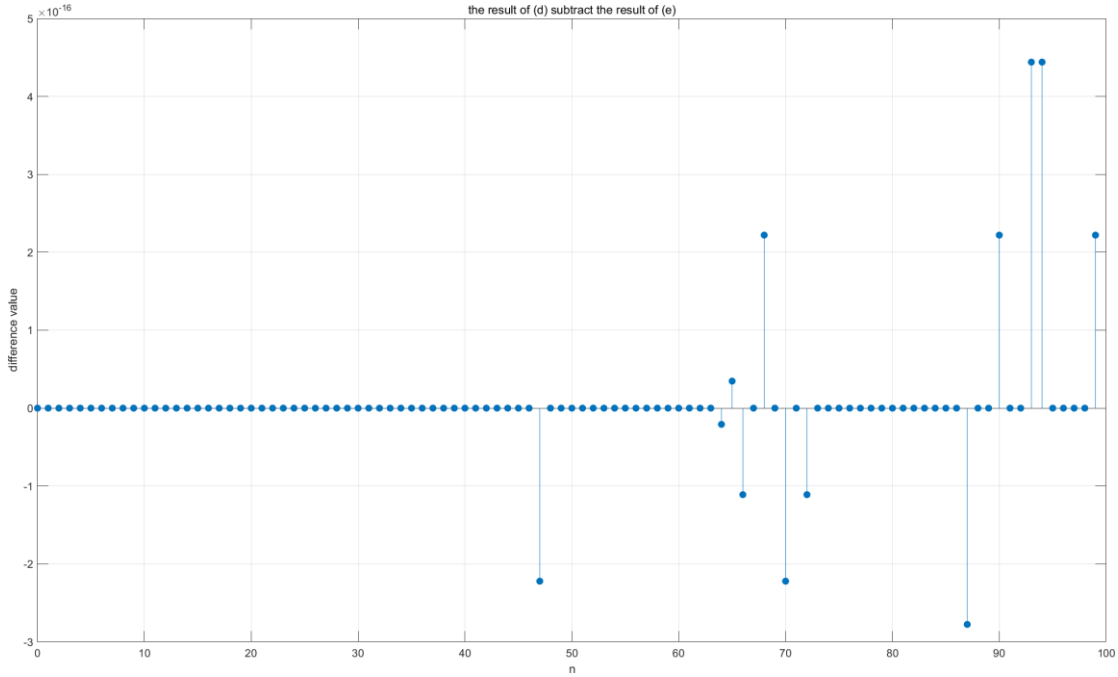
```
nx00=[-9:49];
nx01=[41:99];
ny=[0:99];
x0=[cos(nx00.^2).*sin(2*pi*nx00/5)];
x1=[cos(nx01.^2).*sin(2*pi*nx01/5)];
h=[(0.9).^nh];
y00=conv(h,x0);
y01=conv(h,x1);
y10=[y00 zeros(1,41)];
y11=[zeros(1,40) y01];
y2=[];
for k0=0:49
y2(end+1)=y10(k0+10);
end
y0=[y2 zeros(1,50)];
y3=[];
for k1=50:99
y3(end+1)=y11(k1);
end
y1=[zeros(1,50) y3];
y=y0+y1;
figure(1);
subplot(2,1,1);
stem(ny,y0,'fill');
xlabel('n');
ylabel('y0[n]');
title('The figure of y0[n]');
grid on;
subplot(2,1,2);
stem(ny,y1,'fill');
xlabel('n');
ylabel('y1[n]');
title('The figure of The figure of y1[n]');
grid on;
figure(2);
stem(ny,y,'fill');
xlabel('n');
ylabel('y[n]');
title('The figure of    y[n]=y0[n]+y1[n]');
grid on;
%.....compare the result of (d)and （e） .....%
nx=[-9:99];
x=[cos(nx.^2).*sin(2*pi*nx/5)];
yd0=conv(h,x);
```

```
yd=[];
for k=0:99
yd(end+1)=yd0(k+10);
end
t=yd-y;
figure(3);
stem(ny,t,'fill');
xlabel('n');
ylabel('difference value');
title('the result of (d) subtract the result of (e)');
grid on;
```

figure:







Explanation

According to the figure,it is almost same. The overlap-add method in part (e) successfully reproduced the result from part (d) with minimal numerical error. This method allows efficient convolution computation for long sequences by dividing them into smaller parts, which can be

helpful in practical applications where memory or computation resources are limited.

X0[N]=COS(N^2）$\times \sin\frac{2\pi N}{5}$

h[n]=0.9^N

X1[N]=COS(N^2）$\times \sin\frac{2\pi N}{5}$

When n<40,Y[N]=Y0[N]+0=$\sum_0^{10} x1[N-K]\times h[k]$=$0.9^k \times \mathrm{COS}((N-k)^2\text{）}\times \sin\frac{2\pi(N-K)}{5}$

When 40 <n<60,Y[N]=Y0[N]+Y1[N]= $\sum_0^{10} x1[N-K]\times h[k]$+$\sum_0^{10} x1[N-K]\times h[k]$

When n>60,Y[N]=0+Y1[N]=$\sum_0^{10} x1[N-K]\times h[k]$=$0.9^k \times \mathrm{COS}((N-k)^2\text{）}\times \sin\frac{2\pi(N-K)}{5}$

十、 Summary and comments：

Through this MATLAB lab experiment, I gained a deeper understanding of several important properties of convolution, especially in the context of linear time-invariant (LTI) systems. First, I observed that for time-invariant systems, convolution accurately represents the system's response, producing consistent results whether calculated directly or through iterative methods. However, for time-varying systems, convolution may not capture all dynamic changes, as it inherently assumes time invariance. Additionally, I learned about the **overlap-add method** as an efficient way to compute the convolution of long sequences by breaking them into smaller segments. This method, by dividing the input signal into blocks, convolving each block separately, and then combining the results, maintains accuracy while reducing computational load. Overall, this experiment demonstrated how convolution techniques can be adapted based on system properties and computational efficiency requirements, enhancing my understanding of convolution’s application in practical signal processing.

十、Suggestion for this lab：

None.

core：

Instructor：Aynul Islam